



Course Code: CSE475
Course Title: Machine Learning

Section: 06

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Project Task: 01
Project Title: Explainable ML model for Skin-Cancer Detection

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Title: Exploratory Data Analysis of the HAM10000 skin cancer dataset.

1. Introduction: Exploratory Data Analysis (EDA) is an important step before developing any machine learning or deep learning model. It helps understand the dataset, identify missing values, analyze data distributions, detect class imbalance, and gain insights that guide preprocessing and model development.

In this project, EDA was performed on the **HAM10000 (Human Against Machine with 10000 training images)** dataset, which contains dermoscopic images of skin lesions belonging to seven different disease categories.

2. Dataset Description: The HAM10000 dataset contains **10,015 dermoscopic images** of pigmented skin lesions along with metadata describing each image.

Dataset Information:

Feature	Description
Total Images	10,015
Number of Classes	7
Metadata File	HAM1000_metadata.csv
Image Format	JPG
Image Type	RGB

Metadata Attributes:

- Lesion ID
- Image ID
- Diagnosis (dx)
- Diagnosis Type (dx_type)
- Age
- Sex
- Localization

	lesion_id	image_id	dx	dx_type	age	sex	localization
0	HAM_0000118	ISIC_0027419	bkl	histo	80.0	male	scalp
1	HAM_0000118	ISIC_0025030	bkl	histo	80.0	male	scalp
2	HAM_0002730	ISIC_0026769	bkl	histo	80.0	male	scalp
3	HAM_0002730	ISIC_0025661	bkl	histo	80.0	male	scalp
4	HAM_0001466	ISIC_0031633	bkl	histo	75.0	male	ear

3. Objectives of the EDA: The main objectives of this analysis were:

- To understand the structure of the dataset.
- To identify missing values.
- To analyze the distribution of disease classes.
- To examine patient demographics such as age and gender.
- To study lesion locations on the human body.
- To inspect sample demoscopic images.
- To verify image dimensions before preprocessing.

4. Data Loading and Inspection: The dataset was loaded using the Pandas library. Basic information such as the number of rows, columns, data types, and descriptive statistics was examined.

The dataset contains both categorical and numerical attributes. Most columns were complete, although a small number of missing values were observed in the **Age** column.

5. Missing Value Analysis: Missing values were checked using the `isnull()` function. We observed:

- The **Age** column contains some missing values.
- All other metadata fields are largely complete.

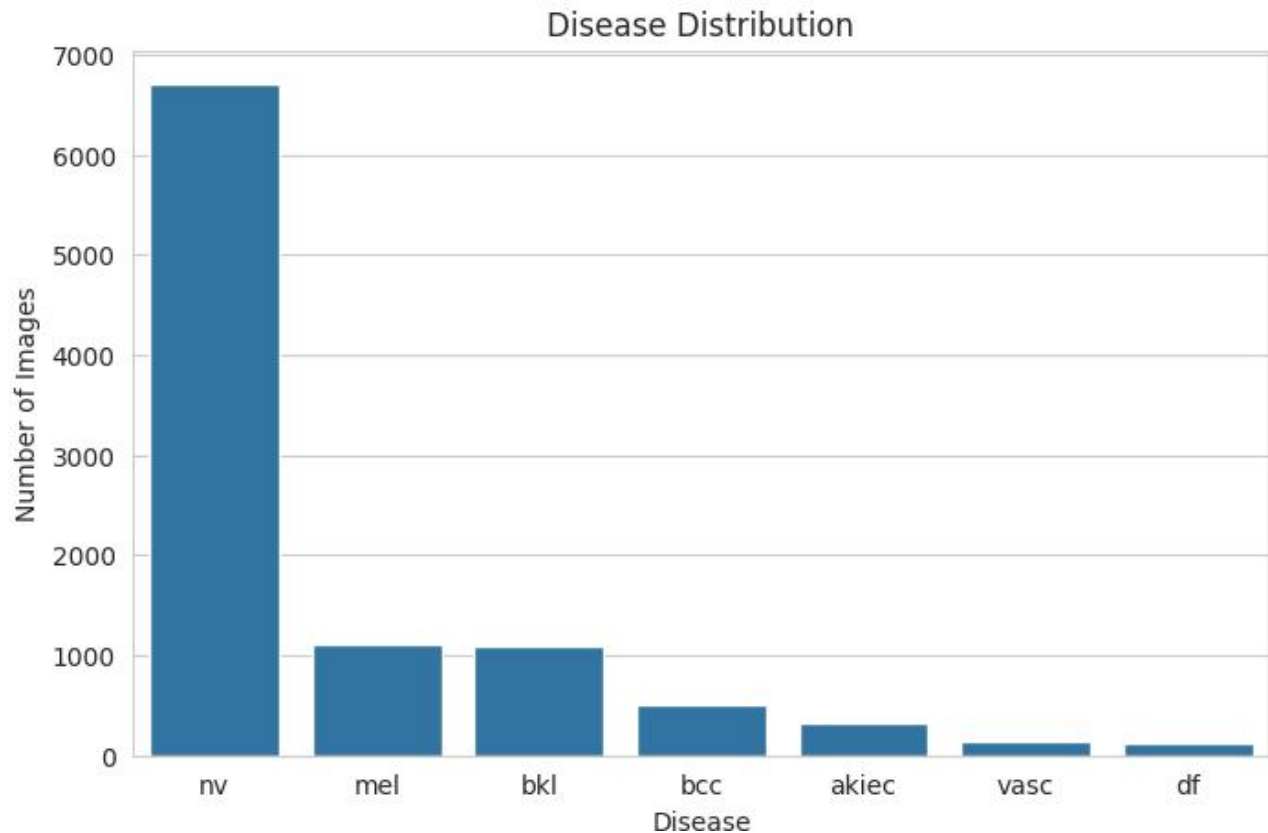


Handling missing values is necessary before model training to avoid inconsistencies.

6. Disease Class Distribution: The frequency of each disease class was visualized using a count plot.

The seven disease classes are:

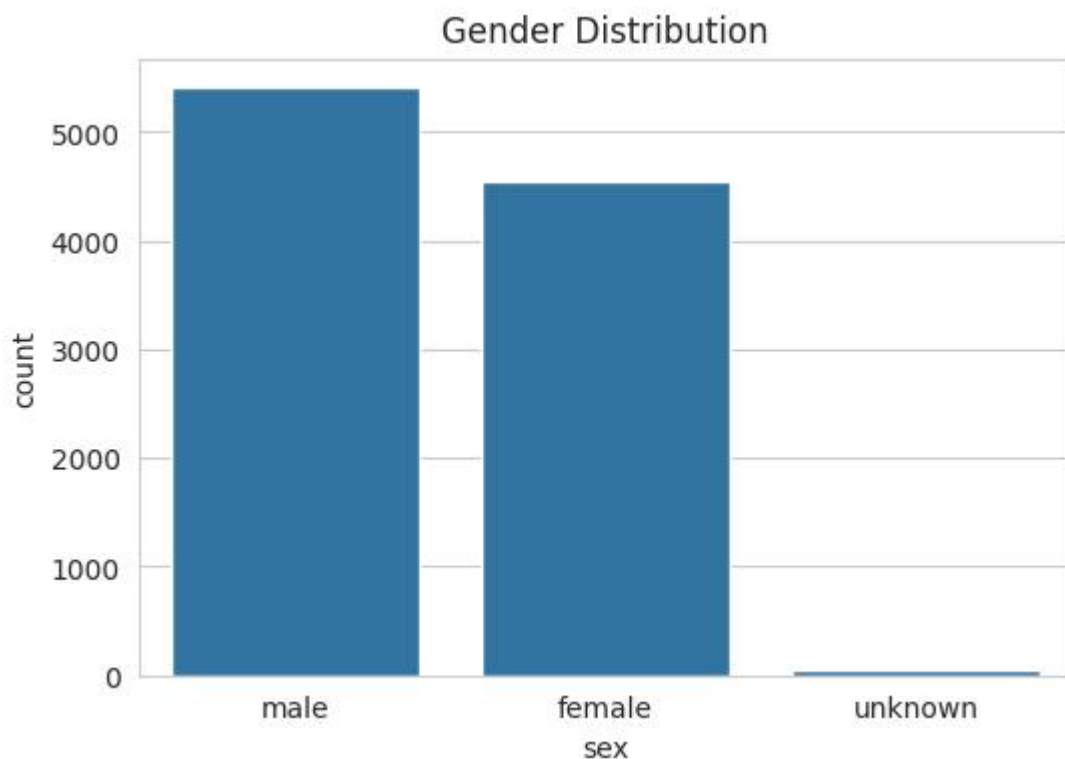
- Actinic Keratoses (akiec)
- Basal Cell Carcinoma (bcc)
- Benign Keratosis (bkl)
- Dermatofibroma (df)
- Melanoma (mel)
- Melanocytic Nevi (nv)
- Vascular Lesions (vasc)



Observation: The dataset is **highly imbalanced**. The **Melanocytic Nevi (nv)** class contains the largest number of images, while **Dermatofibroma (df)** and **Vascular Lesions (vasc)** contain comparatively few samples.

This class imbalance should be considered during model training because it may bias the classifier toward the majority class.

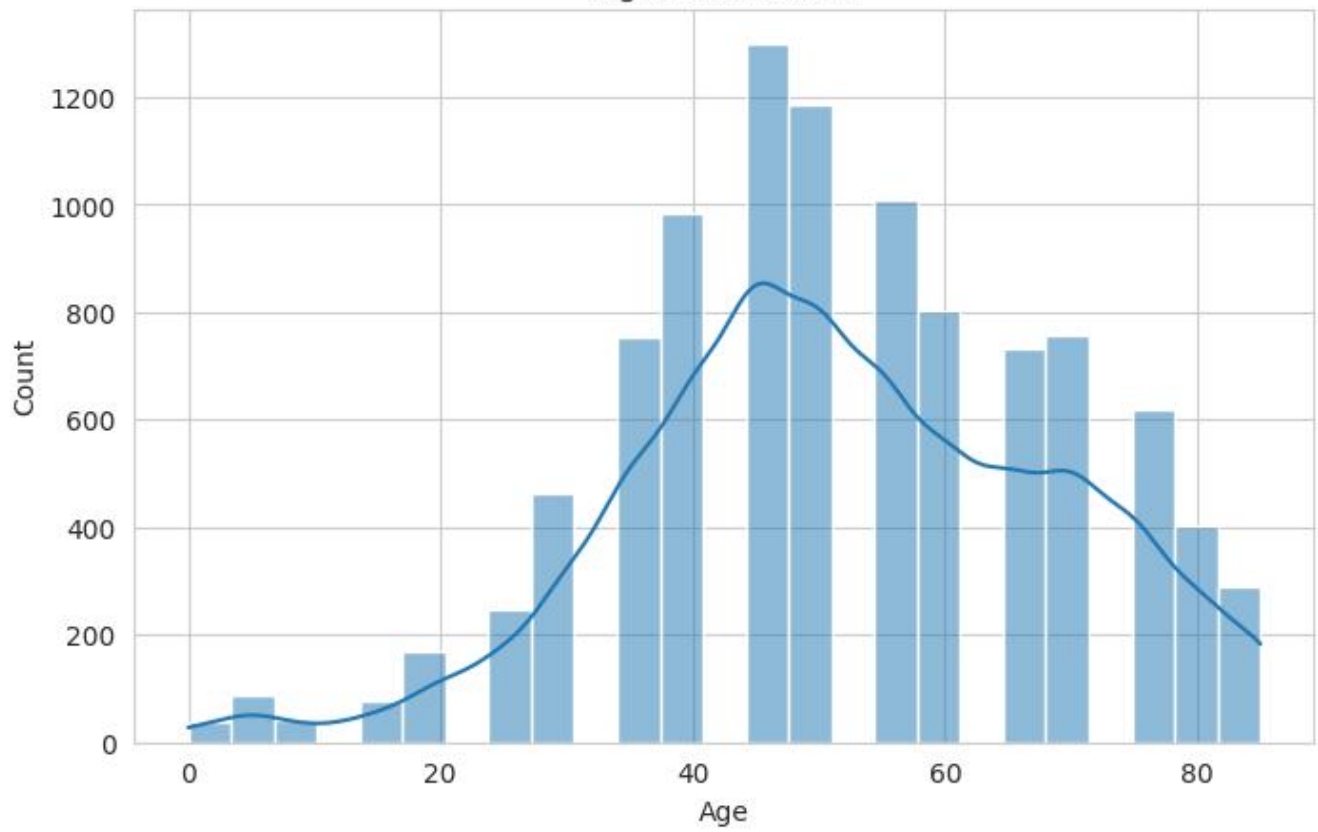
7. Gender Distribution: The gender distribution of patients was analyzed using a count plot.



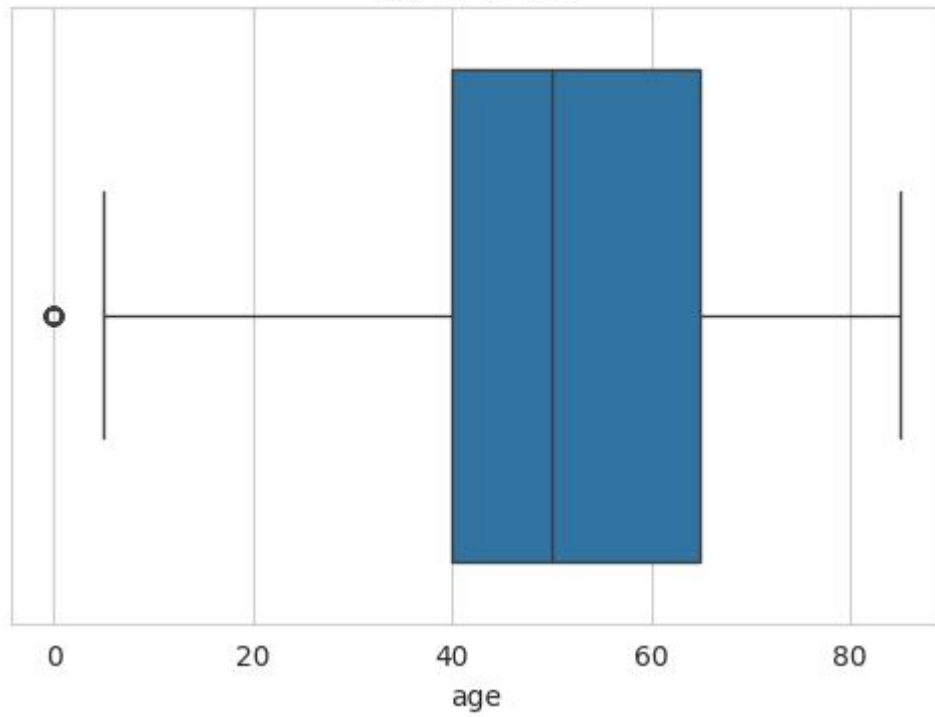
Observation: Both male and female patients are represented in the dataset. The distribution is relatively balanced, although a small number of records have unknown gender information.

8. Age Distribution: The age distribution was visualized using a histogram. Most patients are between 40 and 70 years of age. Very young patients are comparatively rare. This suggests that skin lesions are more frequently recorded among middle-aged and older adults.

Age Distribution



Age Boxplot



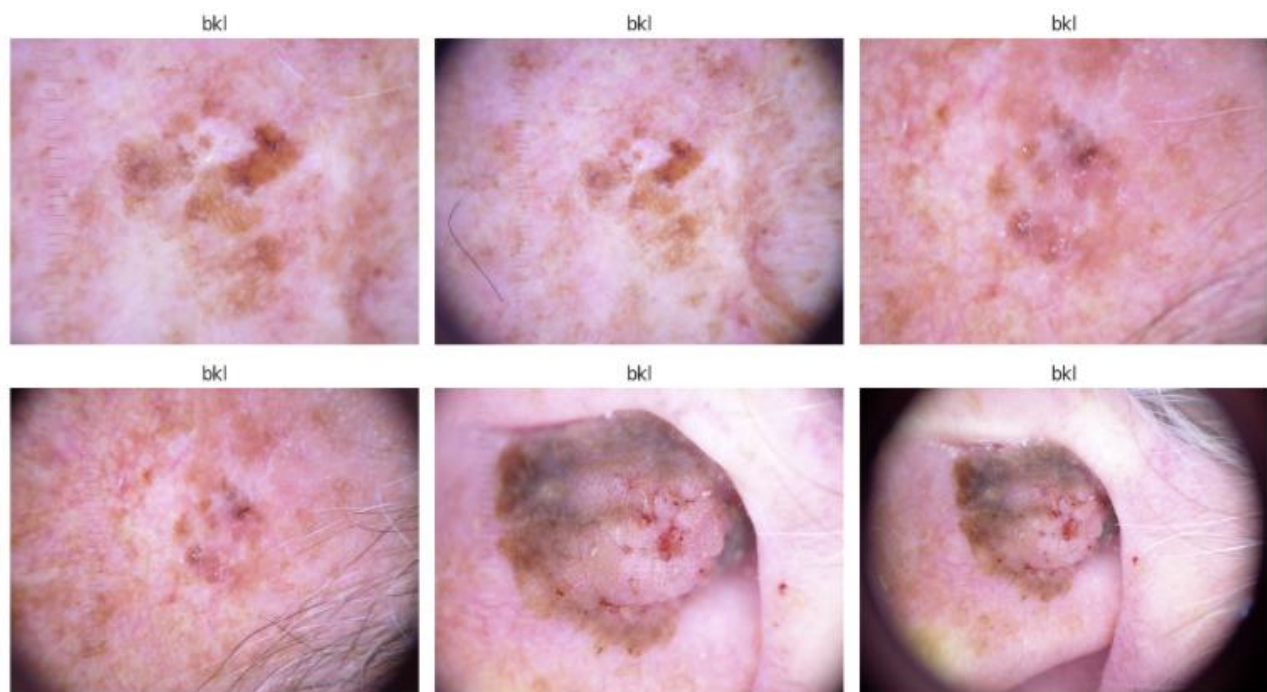
9. Lesion Localization: The body locations of skin lesions were analyzed using a horizontal bar chart. Common lesion locations include: Back, Lower Extremity, Trunk, Upper Extremity. The **back** contains the highest number of recorded lesions, followed by the lower extremities.

10. Diagnosis Type: Different diagnosis methods were analyzed. The metadata includes diagnosis confirmation methods such as:

- Histopathology
- Follow-up
- Consensus
- Confocal Microscopy

Most diagnoses were confirmed using **histopathology**, making the dataset reliable for supervised learning tasks.

11. Sample Image Visualization: Several dermoscopic images were displayed with their corresponding disease labels (ID = ISIC_0024306 to ISIC_0024311).



The above sample images show noticeable differences in color, texture, size, and lesion shape. However, visual similarities between some disease categories indicate that automatic classification is a challenging task.

12. Image Size Analysis: The dimensions of all images were extracted and summarized. We observed that the images have consistent dimensions throughout the dataset. Uniform image sizes simplify preprocessing because additional resizing requirements are minimal.

13. **Key Findings:** The exploratory analysis revealed several important characteristics of the dataset:

- The dataset contains **10,015 dermoscopic images** across **seven disease classes**.
- A small number of missing values exist in the **Age** column.
- The dataset is significantly **class imbalanced**, with the **Melanocytic Nevi (nv)** class dominating.
- Most patients are middle-aged or older.
- Skin lesions most frequently occur on the back and lower extremities.
- Images have consistent dimensions, making preprocessing straightforward.
- Histopathology is the most common diagnosis method, increasing the credibility of the dataset

14. **Conclusion:** The Exploratory Data Analysis provided a comprehensive understanding of the HAM10000 dataset and identified several important characteristics relevant to skin cancer classification. The analysis highlighted class imbalance, missing values, patient demographics, lesion locations, and image properties. These findings will guide the subsequent preprocessing, data augmentation, and model development stages. Overall, the dataset is well-suited for developing deep learning models for automated skin lesion classification, provided that appropriate techniques are applied to address class imbalance and optimize model performance.